Experiment No.: 6

Interfacing of LPC2148 with UART for Serial Communication.

Name: ______________________________

Roll No.: _________ *Batch:* __________

Date of Performance: __________________

Date of Assessment: ____________________

Particulars	Marks
Marks for Regularity (05)	
Marks for Performance (15)	
Understanding (Viva) (05)	
Total (25)	
Signature of Course Teacher	

Experiment: 6

AIM: Interfacing of LPC2148 with UART for Serial Communication.

OBJECTIVE:

1. To interface the LPC2148 microcontroller with a PC using the onboard UART (Universal Asynchronous Receiver/Transmitter) and implement serial data communication through firmware in Embedded C.
2. To understand the working principle of asynchronous serial communication – including frame structure and the role of baud rate in synchronizing transmitter and receiver.

APPARATUS:

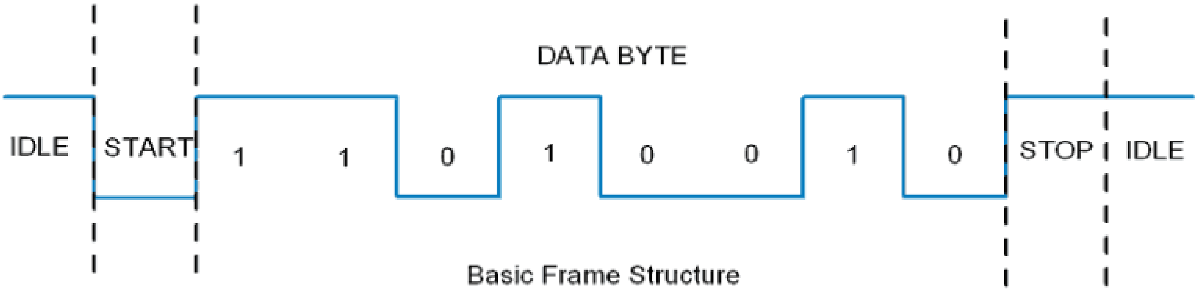
1. LPC2148 module development board.
2. Connecting wire.
3. Adapter.

THEORY:

UART (Universal Asynchronous Receiver/Transmitter) is a serial communication protocol in which data is transferred serially bit by bit at a time. Asynchronous serial communication is widely used for byte-oriented transmission. In Asynchronous serial communication, a byte of data is transferred at a time.
UART serial communication protocol uses a defined frame structure for their data bytes. Frame structure in Asynchronous communication consists:

- **START bit:** It is a bit with which indicates that serial communication has started and it is always low.
- **Data bits packet**: Data bits can be packets of 5 to 9 bits. Normally we use 8-bit data packet, which is always sent after the START bit.

STOP bit: This usually is one or two bits in length. It is sent after data bits packet to indicate the end of frame. Stop bit is always logic high.



UART Frame structure

Usually, an asynchronous serial communication frame consists of a START bit (1 bit) followed by a data byte (8 bits) and then a STOP bit (1 bit), which forms a 10-bit frame as shown in the figure above. The frame can also consist of 2 STOP bits instead of a single bit, and there can also be a PARITY bit after the STOP bit.

LPC2148 has two inbuilt UARTs available i.e. UART0&UART1.

Features of UART0

- 16-byte Receive and Transmit FIFOs
- Built-in fractional baud rate generator with autobauding capabilities
- Software flow control through TXEN bit in Transmit Enable Register

Features of UART1

- 16-byte Receive and Transmit FIFOs
- Built-in fractional baud rate generator with autobauding capabilities
- Software and hardware flow control implementation possible
- Standard modem interface signals included with flow control (auto-CTS/RTS) fully supported in hardware

UART0 Fractional Divider Register:

The UART0 Fractional Divider Register (U0FDR) controls the clock pre-scaler for the baud rate generation and can be read and written at user's discretion. This pre-scaler takes the APB clock and generates an output clock per specified fractional requirements.

Bit	Function	Description
3:0	DIVADDVAL	Baudrate generation pre-scaler divisor value. If this field is 0, fractional baudrate generator will not impact the UART0 baudrate.
7:4	MULVAL	Baudrate pre-scaler multiplier value. This field must be greater or equal 1 for UART0 to operate properly, regardless of whether the fractional baudrate generator is used or not.
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.

This register controls the clock pre-scaler for the baud rate generation. The reset value of the register keeps the fractional capabilities of UART0 disabled making sure that UART0 is fully software and hardware compatible with UARTs not equipped with this feature.

UART0 baud rate can be calculated as:

$$UART0_{baudrate} = \frac{PCLK}{16 \times (256 \times U0DLM + U0DLL) \times \left(1 + \frac{DivAddVal}{MulVal}\right)}$$

Where PCLK is the peripheral clock, U0DLM and U0DLL are the standard UART0 baud rate divider registers, and DIVADDVAL and MULVAL are UART0 fractional baud rate generator specific parameters.

The value of MULVAL and DIVADDVAL should comply to the following conditions:

1. $0 < \text{MULVAL} < 15$
2. $0 < \text{DIVADDVAL} < 15$

If the U0FDR register value does not comply to these two requests then the fractional divider output is undefined. If DIVADDVAL is zero then the fractional divider is disabled and the clock will not be divided.

The value of the U0FDR should not be modified while transmitting/receiving data or data may be lost or corrupted.

UART0 baud rate calculation:

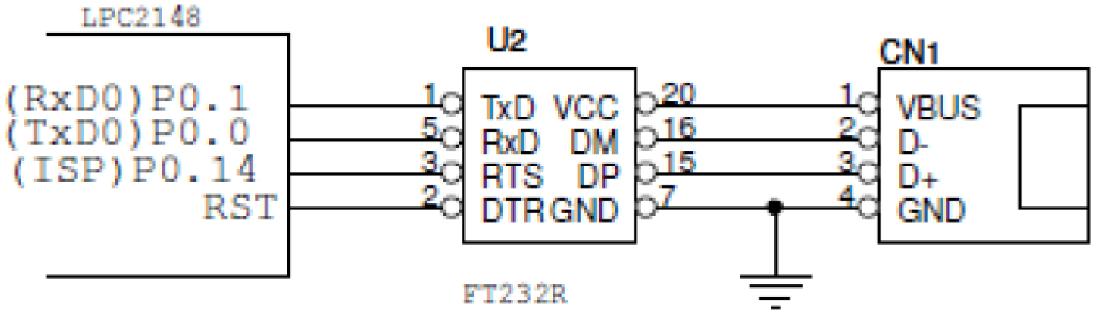
Example 1: Using UART0baudrate formula from above, it can be determined that system with PCLK = 20 MHz, U0DL = 130 (U0DLM = 0x00 and U0DLL = 0x82), DIVADDVAL = 0 and MULVAL = 1 will enable UART0 with UART0baudrate = 9615 bauds.

Example 2: Using UART0baudrate formula from above, it can be determined that system with PCLK = 20 MHz, U0DL = 93 (U0DLM = 0x00 and U0DLL = 0x5D), DIVADDVAL = 2 and MULVAL = 5 will enable UART0 with UART0baudrate = 9600 bauds.

INTERFACING DIAGRAM:

Two UART interfaces from the processor are available for the user on the Micro-A748 board.

UART0 is connected to **USB-to-Serial** on connector **CN1**.



UART-0 interfacing details.

UART0 (USB-to-Serial)DBGU	Pin Details	
	LPC2148	**Description**
RxD0	P0.1	UART0 Receive
TxD0	P0.0	UART0 Transmit

CONCLUSION: